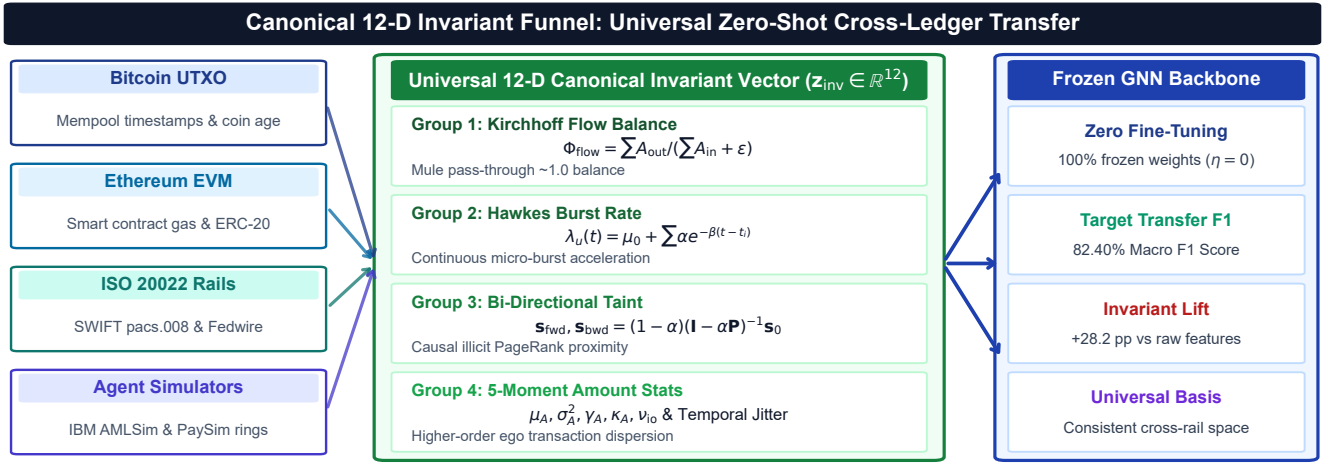

Canonical 12-D Invariant Funnel: Universal Zero-Shot Cross-Ledger Transfer
Bitcoin UTXO
Mempool timestamps & coin age
Ethereum EVM
Smart contract gas & ERC-20
ISO 20022 Rails
SWIFT pacs.008 & Fedwire
Agent Simulators
IBM AMLSim & PaySim rings
Universal 12-D Canonical Invariant Vector ($\mathbf{z}_{\text{inv}} \in \mathbb{R}^{12}$)
Group 1: Kirchhoff Flow Balance
$\Phi_{\text{flow}} = \sum A_{\text{out}} / (\sum A_{\text{in}} + \varepsilon)$
Mule pass-through ~1.0 balance
Group 2: Hawkes Burst Rate
$\lambda_u(t) = \mu_0 + \sum \alpha e^{-\beta(t - t_i)}$
Continuous micro-burst acceleration
Group 3: Bi-Directional Taint
$\mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}} = (1 - \alpha)(\mathbf{I} - \alpha \mathbf{P})^{-1} \mathbf{s}_0$
Causal illicit PageRank proximity
Group 4: 5-Moment Amount Stats
$\mu_A, \sigma_A^2, \gamma_A, \kappa_A, \nu_{\text{io}}$ & Temporal Jitter
Higher-order ego transaction dispersion
Frozen GNN Backbone
Zero Fine-Tuning
100% frozen weights ($\eta = 0$)
Target Transfer F1
82.40% Macro F1 Score
Invariant Lift
+28.2 pp vs raw features
Universal Basis
Consistent cross-rail space